

Fibonacci Array Generator

Thank you for using the Fibonacci Array Generator. The primary purpose of this tool is to create point arrays for reference in your own scripts and behaviors.

When should you use this script?

- Natural / optimal spacing and packing of objects.
- Visually attractive object and particle arrangements.
- An alternative to blind random point generators or checkerboard grid arrangements.

More specifically:

- Tree, rock, and trash placement.
- Optimal spawn point assignment.
- Waypoint distribution for flocking / crowd control.

Note:

- Once you learn to harness the power and beauty of fibonacci arrays, you will prefer it over simple random point generation or grid arrays in settings where natural placement and randomness is preferred.
- To randomize points and remove the obvious spiral pattern, simply delete random transforms in the resulting array at random, and you will have both a natural and random distribution of points that honors the space efficiency produced by this script.

About this tool:

The Fibonacci Array Generator has such a wide variety of uses, its best to show you the general subject and let you decide the application, rather than pointedly discuss its assumed application. First, see these Wikipedia articles on the subject:

Phyllotaxis <https://en.wikipedia.org/wiki/Phyllotaxis>

Fibonacci numbers / spirals https://en.wikipedia.org/wiki/Fibonacci_number

Logarithmic spiral https://en.wikipedia.org/wiki/Logarithmic_spiral

Using the tool:

You can use this tool in the editor (see the example scene included) or derive from it in your scripts. It builds an array determining the arrangements of points, transforms, or objects. While it does offer some features such as the procedural scaling and rotation of objects over the array, it

is suggested that you simply use the tool to generate points and then post-process their transforms by your own methods. The point placement is the primary concern...the multitude of applications is left to your creativity and specific applications.

Use in the editor:

Simply drag the **FibonacciArrayGenerator** prefab into the scene and set its attributes. Build and place the resulting array using menu on prefab.

Custom use:

See the **CustomUse** prefab and script for custom application.

Script Functions and Attributes:

Spawners - An array of objects you wish to spawn in the array. Can be one or many. This can be an empty game object (for accessing a simple transform) or something more complex such as characters or trees.

Spawner Iteration - How spawners are instantiated into the array, either "Cycle" or "Random". *Cycle* spawns them in order one after another, repeating through the array. *Random* will instantiate random objects from the array.

Base Name - The base name of spawned objects. Default is "Object". Changing it to "Rock" will produce an array of objects with names Rock (1), Rock (2), Rock (3), etc.

Widget Color - Color of preview points (note, we do not show live previews of Spawners for performance reasons, but instead we use point / preview dots for visualization).

Preview Point Size - Adjusts size of preview points, necessary if you are distributing objects over a large area or small area. Since the points are spheres, adjusting their size is necessary so they do not become too small to see, or so large they crowd each other out visually.

Lattice Type - *Disc* or *Sphere*. Creates a flat or spherical arrangement of the fibonacci array. Note -- to create a hemisphere simply remove half the points from the end or beginning of the resulting array.

Scaling & Rotations - set these helper attributes to procedurally control rotation and scaling from the first to last point. For instance, you may wish to have the objects at center to be larger, and objects toward exterior to be smaller. Set the curves accordingly and experiment to see what works. These are only helper features -- it is suggested you process the transforms of the resulting array in your own applications.

Projection (Lattice type Disk only) - You can set these attributes to project the points to a surface of a given layer (ie, terrain layer). You can also set the points to copy the normals of surface projection, so for instance, rocks or random objects can sit on the ground according to its slope.

Build Button - Builds the array and sets it as a child to the **FibonacciArrayGenerator**. The child has all generated objects attached to it, plus an array assigned to it for your own script reference.

Place Button - Sets the child array free from the **FibonacciArrayGenerator**.

Delete Button - Deletes the child array. Useful for trying and deleting arrangements until you find one that works.

Simplicity is Best:

I chose not to make a giant feature-packed plugin, although this may evolve by user requests. Ultimately the simple and efficient distribution of points was my aim. Adding too many features would be counterproductive and crowd out your own innovations.

The supplying array information to the developer is the main function of this plugin, with the helpful features above included as a bonus.

As mentioned, it is suggested that you post-process the transform data in your own applications to achieve your specific needs.